

Figure 4: Different components in Wireshark window

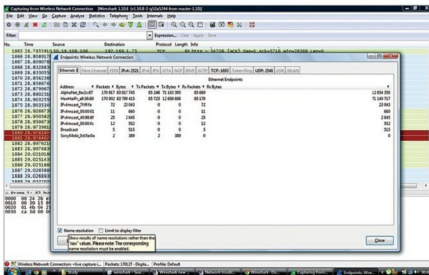


Figure 5: Endpoints display window in Wireshark

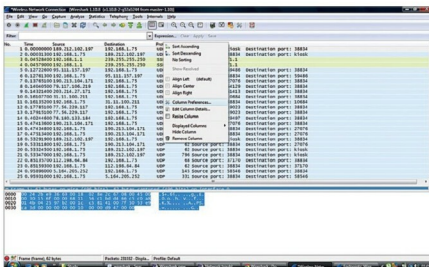


Figure 6: Setting Column Preferences in Wireshark

wireshark.org/download.html.

- Follow the instructions for installation.
- Run the application.
- Select the appropriate network hardware connection you wish to check or trace (for example, local area network or wireless network connection).
- Click **Start** on the home screen (Figure 2) to start capturing live packets from the hardware connection specified.

Figure 4 demonstrates the display filter specification and the list of packets being captured by Wireshark. This information is used for data analysis and for finding a probable error in the communication system.

Details of different Wireshark components and operations are given below.

- On the left-most side of the Wireshark window (Figure 4) we have two parameters: packet number and time. Both parameters are very important when it comes to deciding on the sequence and time of the packet captured. For example, we know that the fault in the system was during x time, say, between 7.30pm and 8.30pm. By changing time characteristics, we can view and study the packets for that period of time only.

To view different timing formats, click **View → Time Display Format**.

- To find out whether the trace you have conducted has devices you are looking for, click **Statistics → Endpoints**. Hence, we can check the source (where the data is coming from) and its destination.

We can also convert the MAC address to names of devices as shown in Figure 5. To see the MAC address, click to turn off the **Name resolution** checkbox at the bottom of the window. You can also view your devices based on their IP address by clicking on the **IPv4 tab**. UDP packets can also be checked.

- You can right-click on **Title bar** of data and change the column